STAFF REVIEW · H446 045 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Use new with a constructor

2. Value held separately per object

3. Reuse and maintainability

4. Last in first out

I Do — source worked steps

1. Constructor sets the initial state — new Account(10) invokes Account(int opening) . It checks 10 is non-negative, then stores 10 in the private field balance .

2. Method validates the proposed change — savings.Deposit(5) passes the positive-amount check and increases balance from 10 to 15. Outside code cannot change the private field directly.

3. Check valid and invalid cases — GetBalance() returns 15. Deposit(-3) throws an exception before changing the balance. Encapsulation is controlled access, not merely writing private .

We Do — guided CHECK

1. 27; invalid deposit rejected
Constructor sets 20; valid deposit makes 27. The second call raises an ArgumentOutOfRangeException before the assignment; when the caller handles it, the balance remains 27 .

2. Apply the same pattern
Identify the precondition before any assignment to width .
Reject invalid initial state
if (w <= 0) throw new ArgumentOutOfRangeException(nameof(w)); width = w; goes inside public Box(int w) . Width 4 is accepted; width 0 is rejected.

You Do — source mark schemes

1. public Box(int w) { width = w; } (with width declared as an instance field).

2. Keeping state inside an object and controlling access through methods/properties.

3. Reject amounts <= 0 before adding to balance.

Exit Ticket — source answer

1. False. In C#, a constructor is a special member used to initialise a new object. It has the class name and no declared return type—not even void.